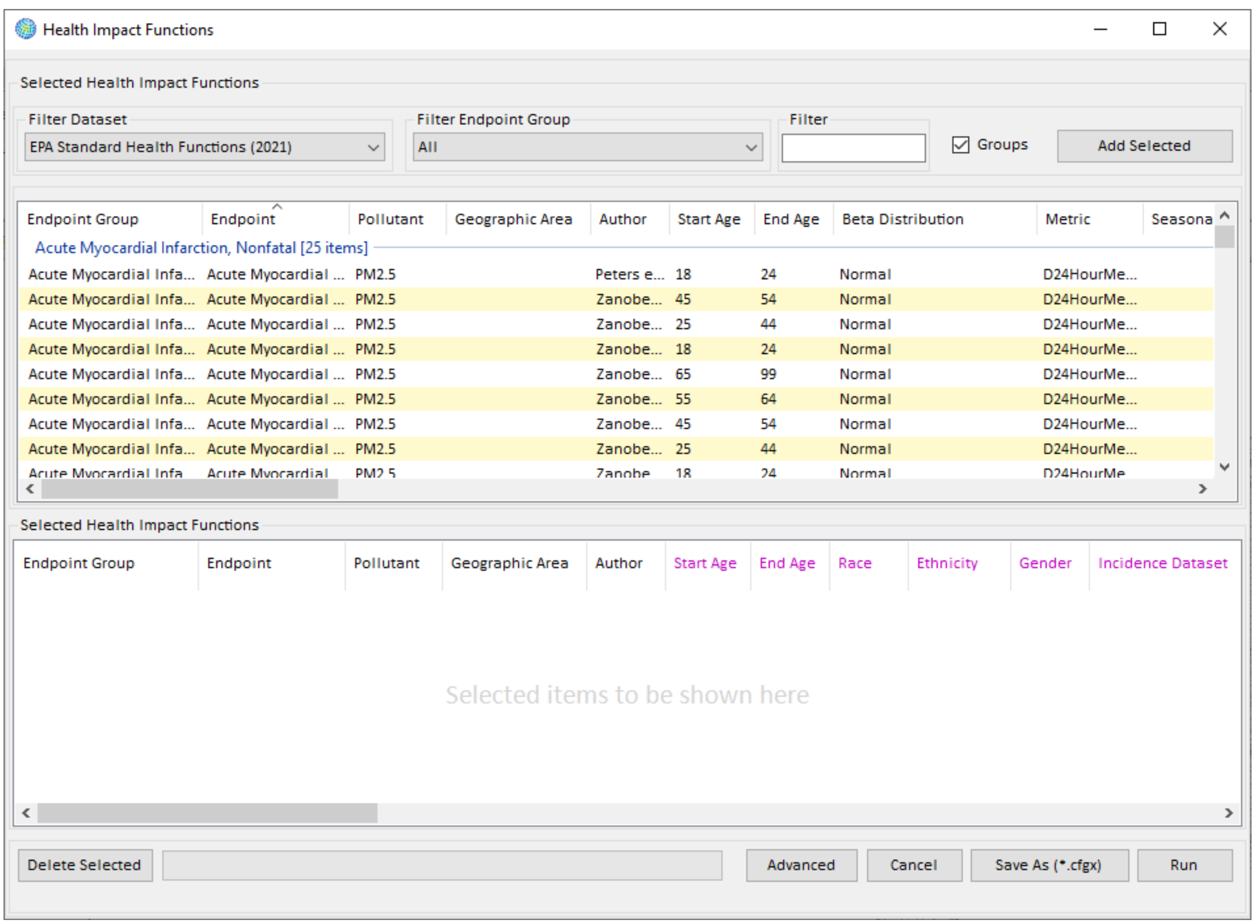
Clicking the **Map** option in the bottom left hand corner will open a separate **Population Map** window. (Warning: this procedure can require a significant amount of time to complete as it creates the population grid and loads the population growth weights). This provides a map of your selected population. If you want to view a different population subgroup based on race, gender, ethnicity, or age range, choose from the available drop-down list (this runs fairly quickly as it uses the same population grid already loaded). The map will refresh automatically.

To change the **Population Dataset** or **Population Year**, click on their respective drop-down list and click **Draw** to update the map and display new results. Warning: This requires BenMAP-CE to recalculate the grid.

Click the close ('×') button at the top of the **Population Map** window to close it.

Clicking **OK** on the **Population Dataset** window will close the window and change the stoplight color to green in the tree menu.

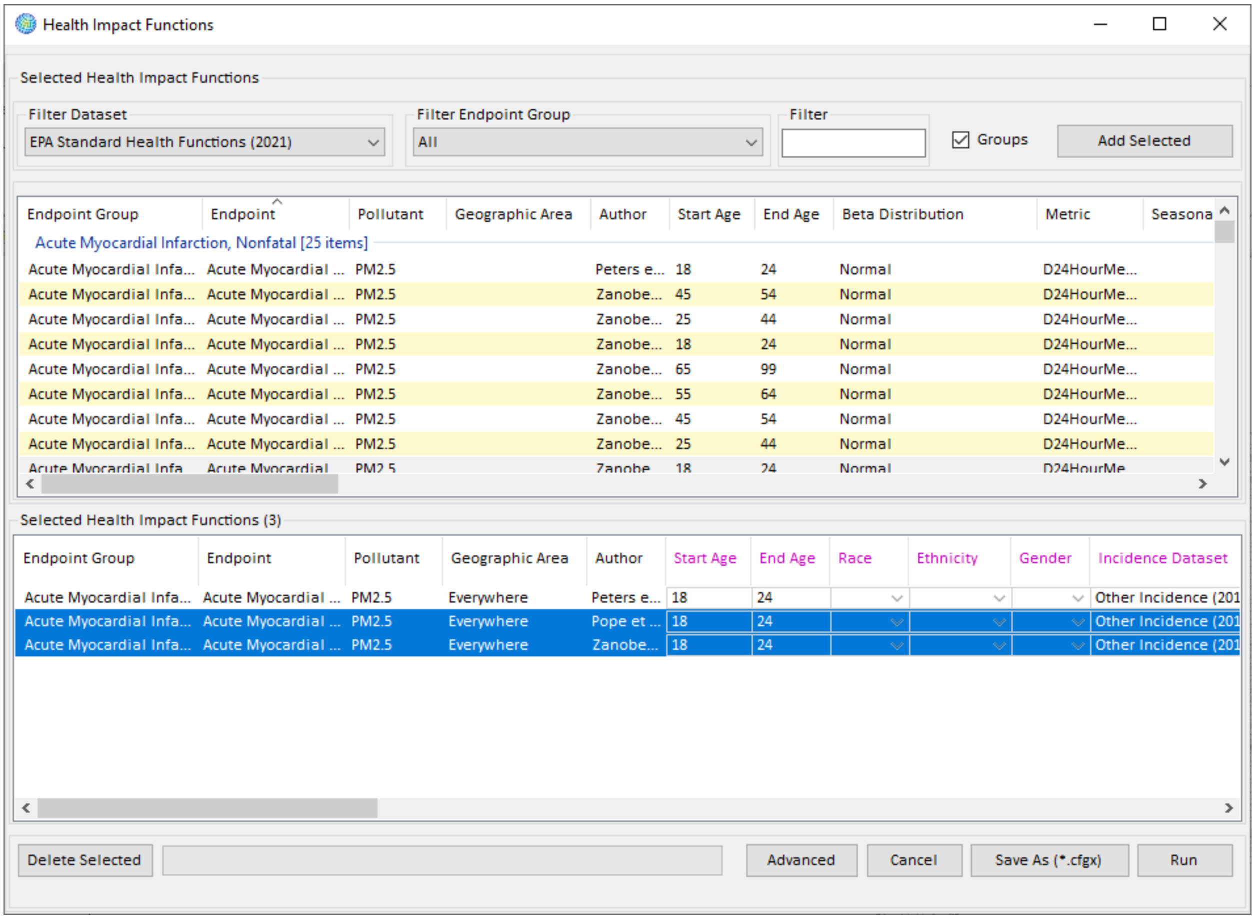
Next, double-click **Health Impact Functions** on the tree menu to display the **Health Impact Functions** window.



The **Health Impact Functions** window is split into two display frames. The upper frame presents the **Available Health Impact Functions**, which you may select, and the lower frame shows the **Selected Health Impact Functions** that you have already selected. Both frames have a tree structure and the ability to change the order of the fields for easy viewing of the functions. Above the **Available Health Impact Functions** frame, there are a series of buttons that give you the option to select different datasets and filter options for each one.

> **BenMAP-CE Guidance and Best Practices**
>
> When selecting HIFs for your analysis, one option is to use the ozone and $PM_{2.5}$ configurations used by EPA. These are available on the BenMAP-CE website (http://www2.epa.gov/benmap/benmap-community-edition). These functions are derived from the epidemiology literature described in the appendices to this user manual.
>
> If BenMAP-CE does not have pre-installed functions for a pollutant or a health impact that you would like to study, you may need to develop your own health impact function(s) which involves a careful review of the epidemiological literature. We recommend consulting with an epidemiologist or other public health professional if you need to develop your own functions.

To add studies to your configuration, simply click to select the health impact functions of interest in the top frame and drag them down to the lower frame. You can do this for blocks of health impact functions by selecting the *Groups* option, clicking on the header of an endpoint group, and dragging the entire endpoint group into the lower frame. And, you can use the options to **Filter Dataset**, **Filter Endpoint Group**, and **Filter** (by keyword) to filter the list according to your preference. Once you are satisfied with the filter, click the **Add Selected** button to apply the selection to the list of selected functions.



If you want to delete some of the health impact functions that you added to your configuration, just highlight the studies of interest and hit the **Delete** key on your keyboard (or click the **Delete Selected** button on the form).

BenMAP-CE displays information for selected studies in two broad categories: **Function Identification** (column headings in black text) and **Function Parameters** (column headings in pink text). The **Function Identification** includes information such as the **Endpoint Group**, **Endpoint**, **Metric**, **Location**, and other variables. This identification information is useful when distinguishing between multiple health impact functions. The **Function Parameters** include those variables that you may directly edit: **Race**, **Ethnicity**, **Gender**, **Start Age**, **End Age**, **Geographic Area, Incidence Dataset**, **Prevalence Dataset**, and **Variable Dataset.**

Fundamental Concepts – Incidence, Prevalence, and Variable Datasets

The **Incidence Dataset** contains **baseline incidence rates,** or the average number of health outcomes (e.g., number of hospital visits or deaths) per person, per unit of time (generally a day or a year)—from air pollution as well as all other causes. These rates are used to calculate the baseline number of adverse health effects that are occurring in a population.

The **Prevalence Dataset** contains **prevalence rates,** or the percentage of individuals in a given population who already have a given adverse health condition. Prevalence rates are used to calculate changes in health conditions among those who already have a health condition, such as asthmatics experiencing asthma symptoms exacerbated by air pollution.

The **Variable Dataset** contains sociodemographic and economic data not contained in other setup datasets that may be used in health impact functions and valuation functions. Examples of socioeconomic variables that might be included in **Variable Data** include wages (including fringe benefits and overhead) and percentage of population living below the poverty line. Variables included in the dataset must be associated with a particular **Grid Definition**).

Clicking on a column header will sort the health impact functions according to that variable. For example, clicking the **Start Age** column will sort the functions by youngest to oldest **Start Age**. You can add the same study to the selection multiple times, and then make edits, in order to be able to calculate the impact of changes in these variables of the function. For example, you can perform an analysis using multiple versions of the same function with different age ranges. To edit the default **Start Age** and **End Age**, just highlight the appropriate cell and type in the desired age values. Keep in mind that these ages represent inclusive age bounds, so if you type in '*10*' and '*12*' this will include all children ages ten, eleven, and twelve years old. If you want to apply a single age year (e.g., only children who are eleven years old), then type the same year (e.g., '*11*') in both the **Start Age** and the **End Age**. Note that the accuracy of the populations calculated for these age ranges will depend on the specificity of the population data present in your selected population dataset. Use the drop-down lists in the **Incidence Dataset**, **Prevalence Dataset**, and **Variable Dataset** fields.

Warning

After selecting your health impact functions, Check the *Incidence Dataset* column in the bottom window and make sure that the correct incidence dataset is selected for each health impact function. This is a step which is often missed and can result in errors. Recall that in order to use a baseline incidence rate with a particular health impact function, the endpoint group and endpoint of the baseline incidence rate must match the endpoint group and endpoint of the health impact function exactly.

Added in version 1.4, the **Geographic Area** field allows users to assign health impact functions to either the entire area of analysis ("Everywhere"), to a specific subset of that area (defined by a specific grid definition), or to all areas outside of one or more specified geographic areas ("Elsewhere"). Note that if an air quality cell intersects multiple geographic areas, BenMAP-CE will calculate health impacts for the geographic area containing the majority of the air quality cell. To access these options, select the drop-down list under the **Geographic Area** header. Please note that users must specify whether grid definitions can be linked to health impact functions when defining new grid definitions or modifying existing grid definitions.